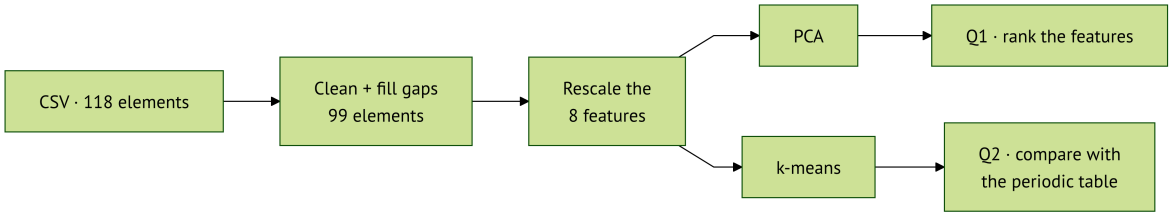
Periodic table data: PCA feature selection and k-means

We have **118 chemical elements** and **8 numeric features** for each one. We answer two questions:

- **Q1 · Which features matter most?** We rank the 8 features with the PCA method of Song, Guo & Mei (2010).
- **Q2 · Do natural groups in the data match the periodic table?** k-means groups the elements by their features only. Then we check if the groups sit together in the table.

The notebook is live. Drag a number, click a chart or drag a list, and the results below update.



Elements analysed
99
19 superheavy elements removed

Most important feature (Q1)
MeltingPoint
top 3 PCA directions

Table neighbours in the same cluster (Q2)
73%
20% expected by chance

New here? Press the button for a short guided tour.

Start Tour

Part 0 · The data

The file has one row for each of the 118 elements. We use these 8 numeric features:

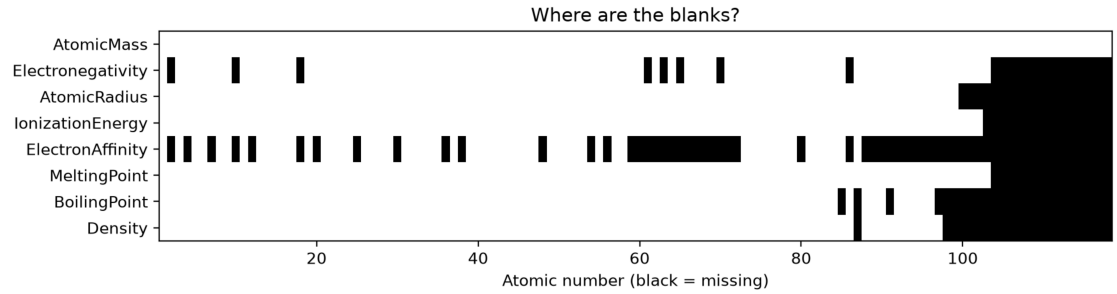
Feature	Unit	Meaning
AtomicMass	u	mass of one atom
Electronegativity	Pauling scale	how strongly the atom pulls shared electrons
AtomicRadius	pm	size of the atom
IonizationEnergy	eV	energy to remove one electron
ElectronAffinity	eV	energy released when the atom takes one extra electron
MeltingPoint	K	temperature where the solid melts
BoilingPoint	K	temperature where the liquid boils
Density	g/cm³	mass per volume

The data has three problems: many heavy elements have **blanks**, `YearDiscovered` contains the text **"Ancient"**, and `StandardState` contains values such as **"Expected to be a Solid"**. We handle each problem below.

	AtomicNumber	Symbol	Name	AtomicMass	CPKHexColor	ElectronConfigura
0	1	H	Hydrogen	1.008000	FFFFFF	
1	2	He	Helium	4.002600	D9FFFF	
2	3	Li	Lithium	7.000000	CC80FF	[He
3	4	Be	Beryllium	9.012183	C2FF00	[He
4	5	B	Boron	10.810000	FFB5B5	[He]2s2
...	...	...	...	...	...	
113	114	Fl	Flerovium	290.192000	NaN	[Rn]7s2 7p2 5f14 6 (predic
114	115	Mc	Moscovium	290.196000	NaN	[Rn]7s2 7p3 5f14 6 (predic
115	116	Lv	Livermorium	293.205000	NaN	[Rn]7s2 7p4 5f14 6 (predic
116	117	Ts	Tennessine	294.211000	NaN	[Rn]7s2 7p5 5f14 6 (predic
117	118	Og	Oganesson	295.216000	NaN	[Rn]7s2 7p6 5f14 6 (predic

118 rows × 17 columns

Only **55 of 118** elements have all 8 features. Most blanks are in the heavy elements (right side). Electron affinity has the most blanks.



	missing	missing_%
ElectronAffinity	61	51.7
BoilingPoint	25	21.2
Electronegativity	23	19.5
Density	22	18.6
AtomicRadius	19	16.1
IonizationEnergy	16	13.6
MeltingPoint	15	12.7
AtomicMass	0	0.0

- Text in `YearDiscovered`: `['Ancient']` (13 elements)
- Values of `StandardState`: `{'Solid': 96, 'Gas': 11, 'Expected to be a Solid': 8, 'Liquid': 2, 'Expected to be a Gas': 1}`

What we do with each problem

Problem	What we do	Why
`YearDiscovered` = "Ancient"	Set a flag `DiscoveredAncient = True`. Leave the year empty.	"Ancient" is a fact (the element was known in old times). It is not a missing value.
`StandardState` = "Expected to be a Solid"	Keep "Solid". Set a flag `StatePredicted = True`.	The state of a man-made element is a prediction, not a measurement.
Elements 100–118 (Fm–Og)	Remove them.	Only 1 to 4 of the 8 features are known. Filling in the rest would invent the element.
No electron affinity for noble gases, group 2, Zn, Cd, Hg, N and Mn	Set it to 0.	These atoms do not hold an extra electron, so the true value is 0 or below.
All other blanks	Fill them from the 5 most similar elements (KNN imputation).	Similar elements have similar values. We check these guesses against a reference library below.
Very different units	Rescale each feature to mean 0 and spread 1 (z-score).	Without this, temperatures in kelvin (up to 5869) hide everything else.

`StandardState` and `GroupBlock` are text labels. They are **not** inputs to PCA or k-means. We use `GroupBlock` and the table position only to **check** the clusters in Q2.

- Known since ancient times (13): C, Al, S, Ca, Fe, Cu, As, Ag, Sn, Sb, Au, Hg, Pb
- Predicted state (9): Ds, Rg, Cn, Nh, Fl, Mc, Lv, Ts, Og
- `StandardState` after cleaning: `{'Solid': 104, 'Gas': 12, 'Liquid': 2}`
- Our period and group for every element match the **mendeleev** library.

The rows we changed (ancient or predicted state):

	Symbol	YearDiscovered	DiscoveredAncient	StandardState	StatePredicted	Block
5	C	<NA>	True	Solid	False	p
12	Al	<NA>	True	Solid	False	p
15	S	<NA>	True	Solid	False	p
19	Ca	<NA>	True	Solid	False	s
25	Fe	<NA>	True	Solid	False	d
...	...	...	...	...	...	...
113	Fl	1998	False	Solid	True	p
114	Mc	2003	False	Solid	True	p
115	Lv	2000	False	Solid	True	p
116	Ts	2010	False	Solid	True	p
117	Og	2006	False	Gas	True	p

22 rows × 8 columns

- Removed **19** elements with fewer than 5 known features: Fm, Md, No, Lr, Rf, Db, Sg, Bh, Hs, Mt, Ds, Rg, Cn, Nh, Fl, Mc, Lv, Ts, Og.
- Kept **99 elements** (H to Es).
- Set electron affinity to 0 for 17 elements: He, Be, N, Ne, Mg, Ar, Ca, Mn, Zn, Kr, Sr, Cd, Xe, Ba, Hg, Rn, Ra.
- Filled **42** other blanks with KNN.

Show the filled-in values

Symbol	AtomicMass	Electronegativity	AtomicRadius	IonizationEnergy	ElectronAffinity
He	NaN	3.090	NaN	NaN	NaN
Ne	NaN	2.807	NaN	NaN	NaN
Ar	NaN	2.893	NaN	NaN	NaN
Pr	NaN	NaN	NaN	NaN	0.447
Nd	NaN	NaN	NaN	NaN	0.386
...	...	...	...	...	...
Am	NaN	NaN	NaN	NaN	0.424
Cm	NaN	NaN	NaN	NaN	0.432
Bk	NaN	NaN	NaN	NaN	0.426
Cf	NaN	NaN	NaN	NaN	0.401
Es	NaN	NaN	NaN	NaN	0.401

31 rows × 8 columns

Check: how good are the filled-in values?

The **mendeleev** library has reference values for 13 of the 42 values we filled in. The typical (median) error is **0.61** of the feature's spread (1.0 = one standard deviation). So the guesses are **rough**. Most of the larger errors are in the electron affinity of the lanthanides, where even the reference values are uncertain (many are calculated, not measured). This is one reason we test later if the clusters change when we fill the blanks in other ways, or leave out all filled values. They do not change much. mendeleev also lists a **negative** electron affinity for He, Be, N, Ar, Xe. A negative value means the atom does not hold the extra electron. This supports our choice of 0 for these elements.

Show the comparison

	element	feature	our guess (KNN)	reference (mendeleev)	error / feature spread
0	Pr	ElectronAffinity	0.447	0.962	0.650
1	Nd	ElectronAffinity	0.386	1.916	1.931
2	Eu	ElectronAffinity	0.383	0.864	0.607
3	Tb	ElectronAffinity	0.573	1.165	0.747
4	Dy	ElectronAffinity	0.317	0.352	0.044
...	...	...	...	...	...
8	Hf	ElectronAffinity	0.629	0.014	0.776
9	Fr	Density	3.844	1.870	0.333
10	Ac	ElectronAffinity	0.265	0.350	0.108
11	Cf	Density	15.831	15.100	0.123
12	Es	Density	16.087	8.840	1.224

13 rows × 5 columns

Q1 · Which features matter most?

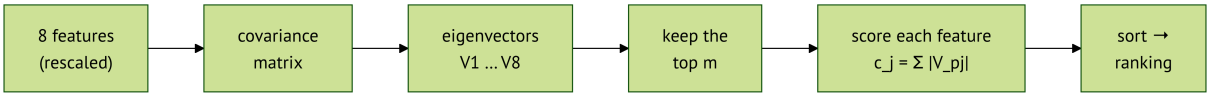
The idea in plain words

PCA finds new **directions** in the data. Each direction is a mix of the 8 features. The first direction holds the most variation between elements, the second holds the next most, and so on.

The **weights** in the mix tell us how much each feature counts in a direction. If a feature has small weights in all the important directions, it adds little. We can remove it and lose

almost nothing.

The method of Song, Guo & Mei (2010)



1. Compute the covariance matrix of the data. Find its eigenvectors (directions) and eigenvalues (how much variation each direction holds).
2. Keep the m eigenvectors with the largest eigenvalues: $V_1, \ldots, V_m$.
3. Give each feature j a score: the sum of its absolute weights in these directions,

$$c_j = \sum_{p=1}^{m} |V_{pj}|$$

4. Sort the scores from high to low. This order is the ranking. To keep n features, keep the top n.

The authors tested the method on face photos, where each pixel is one feature. With a nearest-neighbour classifier, 600 of 2000 pixels gave a **lower** error than all pixels (AR faces: 0.505 → 0.319).

Reference: F. Song, Z. Guo, D. Mei, "Feature selection using principal component analysis", ICSEM 2010, pp. 27–30, doi:10.1109/ICSEM.2010.14.

Two choices we must make. The paper does not say how to pick m. We use the Kaiser rule: keep directions with eigenvalue > 1. The paper also does not rescale the data. Our features have very different units, so we rescale them first (we show why in a "Go deeper" box below).

Before you look: make a guess

Which features do you think matter most? **Drag** them into order (top = most important). We compare your guess with the PCA answer below.

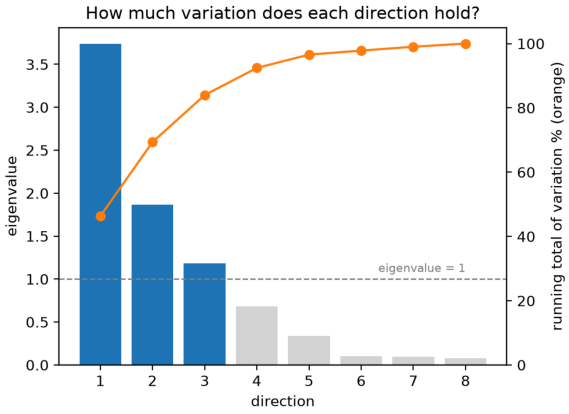
Most important at the top

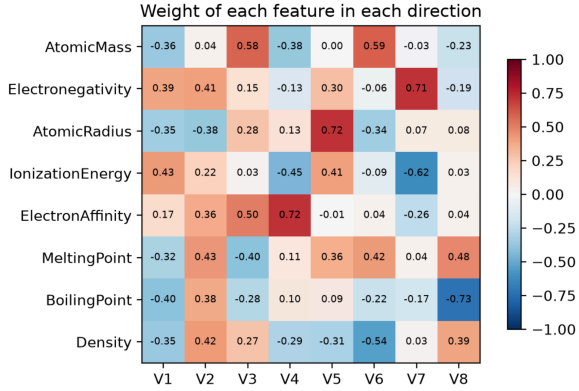
- AtomicMass
- AtomicRadius
- BoilingPoint
- Density
- ElectronAffinity
- Electronegativity
- IonizationEnergy
- MeltingPoint

Show the eigenvalue table

	eigenvalue	explained_%	cumulative_%
PC1	3.741	46.292	46.292
PC2	1.863	23.050	69.342
PC3	1.183	14.632	83.974
PC4	0.683	8.450	92.425
PC5	0.335	4.145	96.569
PC6	0.101	1.251	97.821
PC7	0.097	1.194	99.015
PC8	0.080	0.985	100.000

The first **3** directions have an eigenvalue above 1 (blue bars). Together they hold **84%** of the variation. So our default is **m = 3**. In the right chart, dark red or dark blue means a big weight. The sign does not matter, only the size.



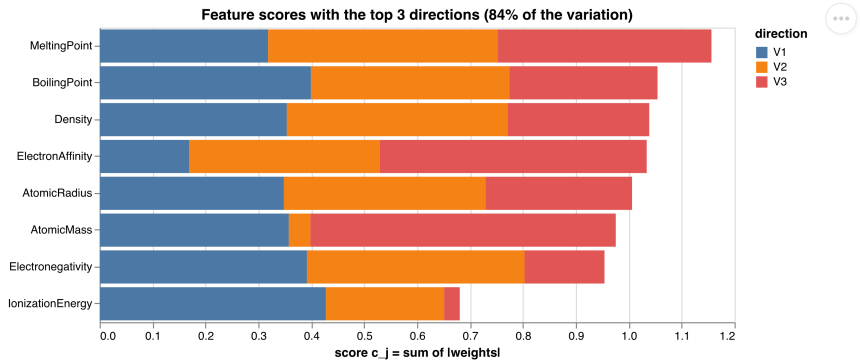


Score the features

Use the top 3 PCA directions to score the features.

Drag the blue number left or right to change m (1 to 8). All charts below update.

Each bar is the score of one feature. The coloured pieces show how much each direction adds. **Hover** over a piece to see the number. Longest bar = most important feature.

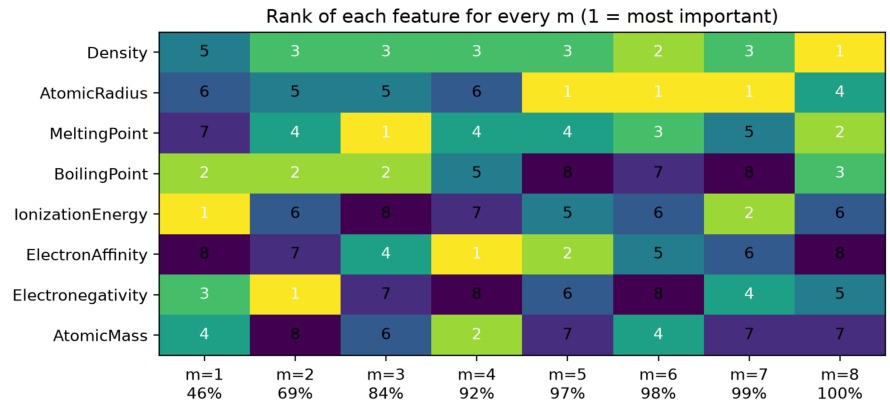


Show the scores as a table

	score c_j	rank	λ-weighted score	λ-weighted rank
MeltingPoint	1.156	1	2.478	2
BoilingPoint	1.054	2	2.524	1
Density	1.039	3	2.417	3
ElectronAffinity	1.034	4	1.901	8
AtomicRadius	1.006	5	2.340	5
AtomicMass	0.975	6	2.096	6
Electronegativity	0.954	7	2.410	4
IonizationEnergy	0.681	8	2.051	7

> Drag the list in **“Make a guess”** above to compare your guess with PCA.

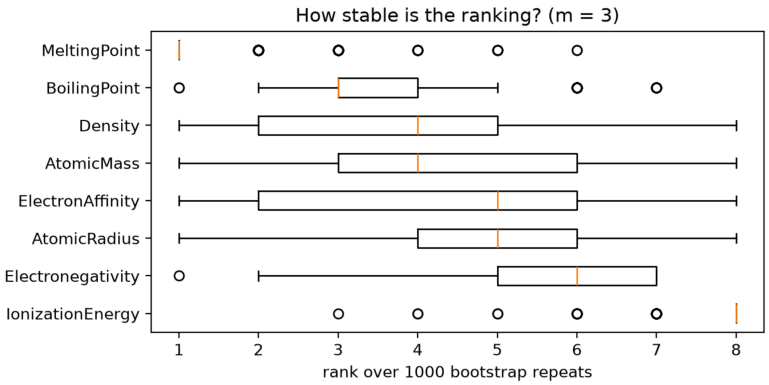
Go deeper · Does the ranking change with m?



Yes, a lot. With only 8 features, each extra direction adds its own features with full weight. With m = 1 the score is just the first direction, which favours ionization energy, boiling point and electronegativity. With m = 8 the score uses all directions and says little.

Go deeper · Is the ranking stable? (bootstrap)

We draw 1000 new data sets by picking elements at random with replacement, and rank the features again each time. A narrow box means a stable rank.



	rank (full data)	bootstrap mean rank	bootstrap rank sd	P(top 3)	P(bottom 3)
MeltingPoint	1	1.34	0.70	0.98	0.00
BoilingPoint	2	3.43	1.14	0.56	0.04
Density	3	3.91	1.72	0.43	0.23
AtomicMass	6	4.28	1.80	0.35	0.29
ElectronAffinity	4	4.35	2.13	0.39	0.36
AtomicRadius	5	4.87	1.59	0.25	0.41
Electronegativity	7	5.90	1.19	0.04	0.68
IonizationEnergy	8	7.92	0.38	0.00	1.00

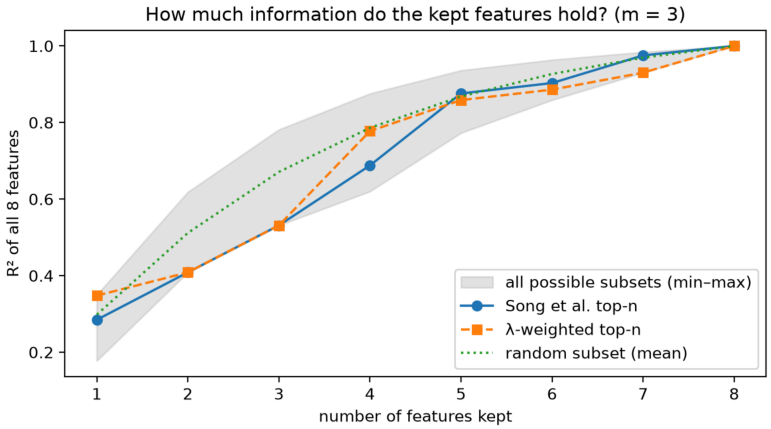
Go deeper · Why we rescale the data first

Without rescaling, the first direction alone holds 95.5% of the variation. It is almost only BoilingPoint (0.88) and MeltingPoint (0.48). These are in kelvin, so their numbers are huge (variance near 10^6). The ranking then shows which feature has the biggest unit, not which feature carries information.

	variance (raw units)	|V1| raw	c_j raw covariance	rank raw	rank rescaled
MeltingPoint	814797.683	0.477	1.411	1	1
BoilingPoint	2429291.843	0.879	1.400	2	2
AtomicMass	5663.625	0.013	1.022	3	6
AtomicRadius	1487.578	0.004	0.356	4	5
Density	35.052	0.002	0.044	5	3
IonizationEnergy	11.427	0.001	0.016	6	8
Electronegativity	0.438	0.000	0.003	7	7
ElectronAffinity	0.628	0.000	0.000	8	4

Go deeper · Do the top features keep the information?

We keep the top n features and try to predict all 8 features from them with a linear model. R^2 = 1 means nothing is lost. The grey band shows the worst and the best of all possible sets of n features.



n features kept	Song et al. top-n	λ-weighted top-n	random subset (mean)	best subset	worst subset
1	0.285	0.348	0.298	0.349	0.178
2	0.408	0.408	0.512	0.618	0.408
3	0.531	0.531	0.671	0.781	0.531
4	0.688	0.778	0.786	0.875	0.620
5	0.876	0.859	0.868	0.936	0.773
6	0.903	0.886	0.927	0.963	0.859
7	0.975	0.930	0.969	0.984	0.930
8	1.000	1.000	1.000	1.000	1.000

Answer to Q1

With the top **3** directions (84% of the variation), the PCA ranking is:

1. **MeltingPoint**
2. **BoilingPoint**
3. **Density**
4. ElectronAffinity
5. AtomicRadius
6. AtomicMass
7. Electronegativity
8. IonizationEnergy

What this means

- **MeltingPoint** is the most important feature. It has a large weight in all 3 top directions.
- **IonizationEnergy** is the least important. It has a large weight in the first direction only. The method adds all 3 directions with the same weight, so this does not help it.
- The top and the bottom of the list are **stable**. In 1000 bootstrap repeats, MeltingPoint stays in the top 3 in 98% of cases, and IonizationEnergy stays in the bottom 3 in 100% of cases.
- The **middle** of the list is not stable. It changes with m and with small changes in the data.

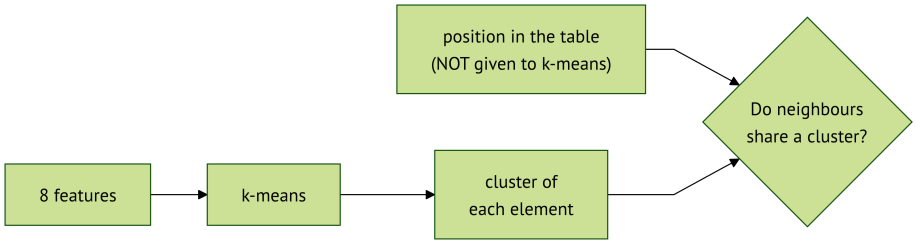
Limit of the method

The method does not see when two features carry the same information. Melting point and boiling point are strongly linked (r = 0.88), so both get a high score. As a result, the top 3 features keep only R^2 = 0.53 of the information in all 8 features. Three random features keep 0.67 on average. So the method is good to find the features that drive the main patterns. It is not good to find the smallest set of features without repeats.

Q2 · Do the clusters match the periodic table?

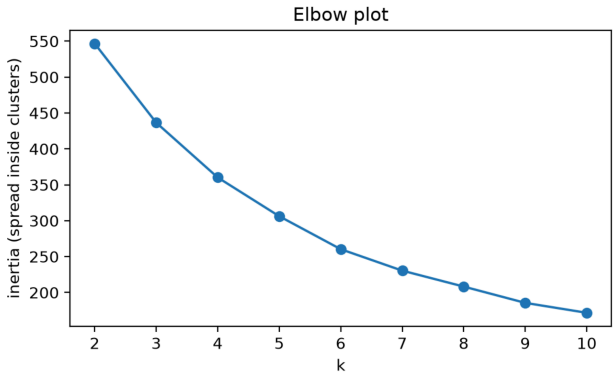
Plan in plain words

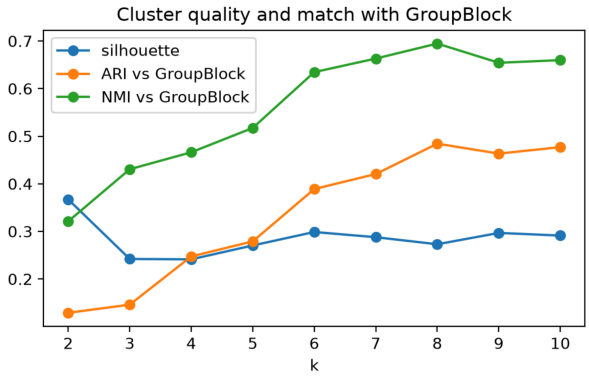
1. **k-means** puts the 99 elements into k groups (clusters), using only the 8 rescaled features. Elements in one cluster have similar features. k-means does **not** know where an element sits in the table.
2. We **colour the periodic table** by cluster and look.
3. We **test** it: are neighbours in the table in the same cluster more often than by chance?



How we picked k

- **Inertia** is the spread inside the clusters. It always falls as k grows; we look for a bend (elbow).
- **Silhouette** (−1 to 1) is high when clusters are tight and far apart.
- **ARI / NMI** show the match with the chemical families in `GroupBlock` (0 = chance, 1 = perfect).





k	inertia	silhouette	ARI vs GroupBlock	NMI vs GroupBlock
2	546.667	0.367	0.129	0.321
3	436.535	0.242	0.146	0.431
4	360.183	0.241	0.247	0.466
5	306.252	0.270	0.279	0.517
6	260.319	0.299	0.389	0.635
7	230.453	0.288	0.420	0.663
8	208.257	0.273	0.484	0.695
9	185.584	0.297	0.463	0.654
10	171.744	0.291	0.477	0.660

Make the clusters

Split the elements into 6 clusters.

Drag the blue number (2 to 10). Everything below updates.

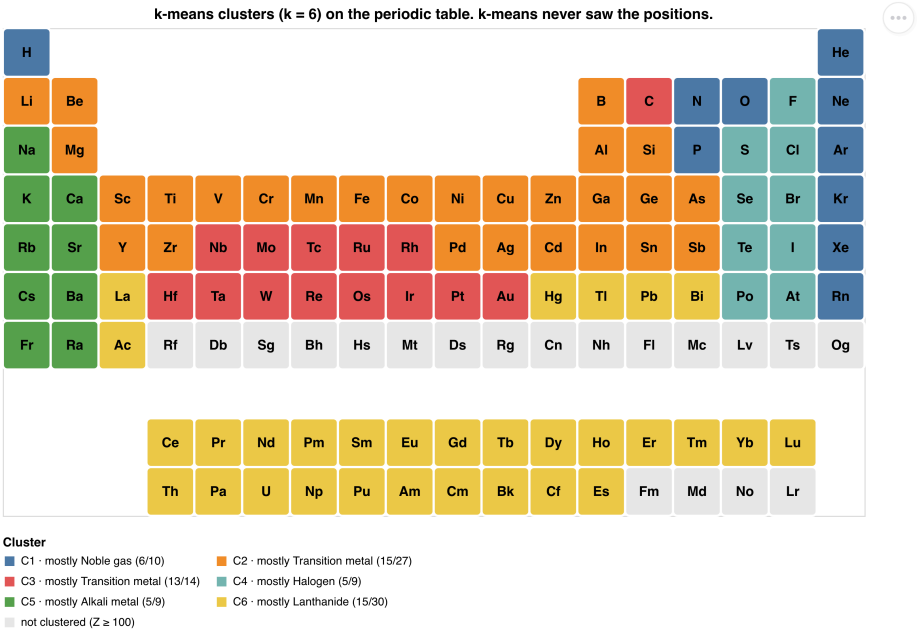
Why 6? k = 2 has the best silhouette (0.37), but it only cuts the elements into two big halves. That is too rough to compare with the table. For k ⩾ 3, the silhouette is highest at k = 6 (0.30), and the elbow plot bends there. The match with the chemical families also grows from ARI 0.13 (k = 2) to 0.39 (k = 6).

Cluster	name	size	members
1	C1 · mostly Noble gas (6/10)	10	H He N O Ne P Ar Kr Xe Rn
2	C2 · mostly Transition metal (15/27)	27	Li Be B Mg Al Si Sc Ti V Cr Mn Fe Co Ni Cu Zn Ga Ge As Y Zr Pd Ag Cd In Sn Sb
3	C3 · mostly Transition metal (13/14)	14	C Nb Mo Tc Ru Rh Hf Ta W Re Os Ir Pt Au
4	C4 · mostly Halogen (5/9)	9	F S Cl Se Br Te I Po At
5	C5 · mostly Alkali metal (5/9)	9	Na K Ca Rb Sr Cs Ba Fr Ra
6	C6 · mostly Lanthanide (15/30)	30	La Ce Pr Nd Pm Sm Eu Gd Tb Dy Ho Er Tm Yb Lu Hg Tl Pb Bi Ac Th Pa U Np Pu Am Cm Bk Cf Es

Show the average features of each cluster

Cluster	C1	C2	C3	C4	C5	C6
size	10.00	27.00	14.00	9.00	9.00	30.00
AtomicMass	56.32	64.40	143.27	102.10	110.50	193.66
Electronegativity	2.74	1.72	2.07	2.69	0.87	1.34
AtomicRadius	172.70	185.96	203.29	185.11	280.78	231.67
IonizationEnergy	15.10	7.52	8.13	11.08	4.86	6.16
ElectronAffinity	0.30	0.56	1.05	2.68	0.27	0.48
MeltingPoint	103.71	1367.51	2792.63	398.29	639.95	1296.48
BoilingPoint	135.65	2742.52	4748.07	720.39	1406.94	3042.65
Density	0.18	5.99	15.15	4.16	2.44	11.04

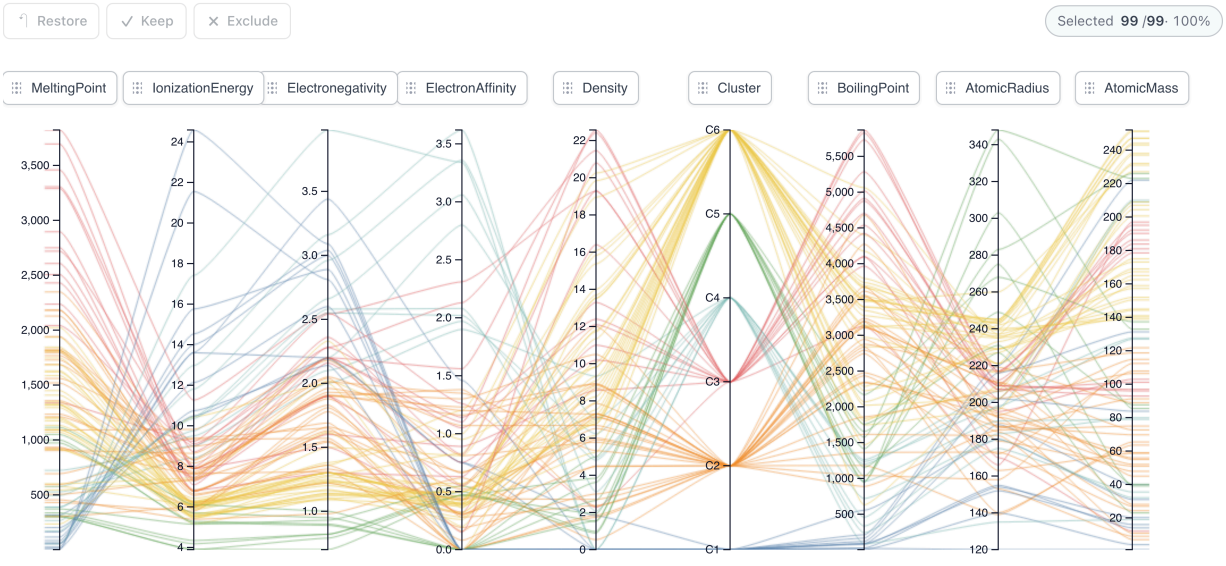
Click an element to light up its cluster (click empty space to reset). **Hover** to see its data. Rows 9–10 are the lanthanides and actinides.



> Click an element in the table above to see the fingerprint of its cluster here.

Cluster fingerprints

Each line is one element. Each vertical axis is one feature (in its own units). Lines of one colour follow a similar path: the elements in a cluster have similar features. **Drag along an axis** to keep only the elements in that range. Cluster names (C1, C2, …) are the same as in the table above.



Do the clusters follow the chemical families?

ARI = 0 means no better than chance, 1 means a perfect match. Each table counts how many elements of each family (rows) fall in each cluster (columns).

	ARI	NMI	n categories
GroupBlock	0.389	0.635	10
Block	0.397	0.539	4
Group	0.424	0.555	19
Period	0.255	0.370	7

Family (GroupBlock) | Block (s/p/d/f) | Group | Period

Cluster GroupBlock	C1 · mostly Noble gas (6/10)	C2 · mostly Transition metal (15/27)	C3 · mostly Transition metal (13/14)	C4 · mostly Halogen (5/9)	C5 · mostly Alkali metal (5/9)	C6 · mostly Lanthanide (15/30)
Actinide	0	0	0	0	0	11
Alkali metal	0	1	0	0	5	0
Alkaline earth metal	0	2	0	0	4	0
Halogen	0	0	0	5	0	0
Lanthanide	0	0	0	0	0	15
Metalloid	0	5	0	2	0	0
Noble gas	6	0	0	0	0	0
Nonmetal	4	0	1	2	0	0
Post-transition metal	0	4	0	0	0	3
Transition metal	0	15	13	0	0	1

The proximity test: are neighbours in the same cluster?

Two elements are **neighbours** when they touch in the table: side by side in a period, or one above the other in a group. We count how many neighbour pairs share a cluster. Then we **shuffle** the cluster labels 5000 times to see what chance gives.

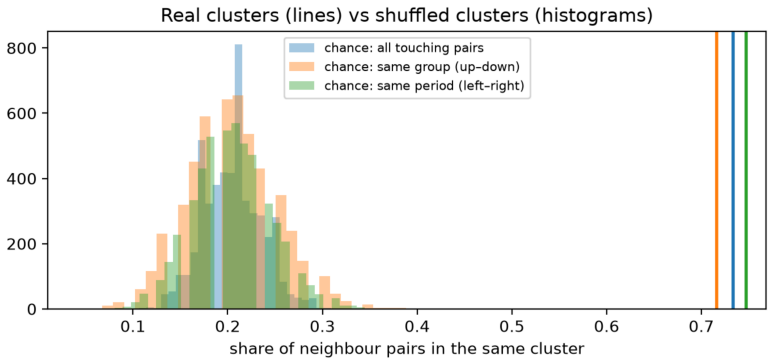
Neighbours in the same cluster
73%
154 neighbour pairs

Expected by chance
20%
cluster labels shuffled 5000 times

p-value
0.0002
no shuffle reached the real value

Go deeper · the full test (up–down vs left–right)

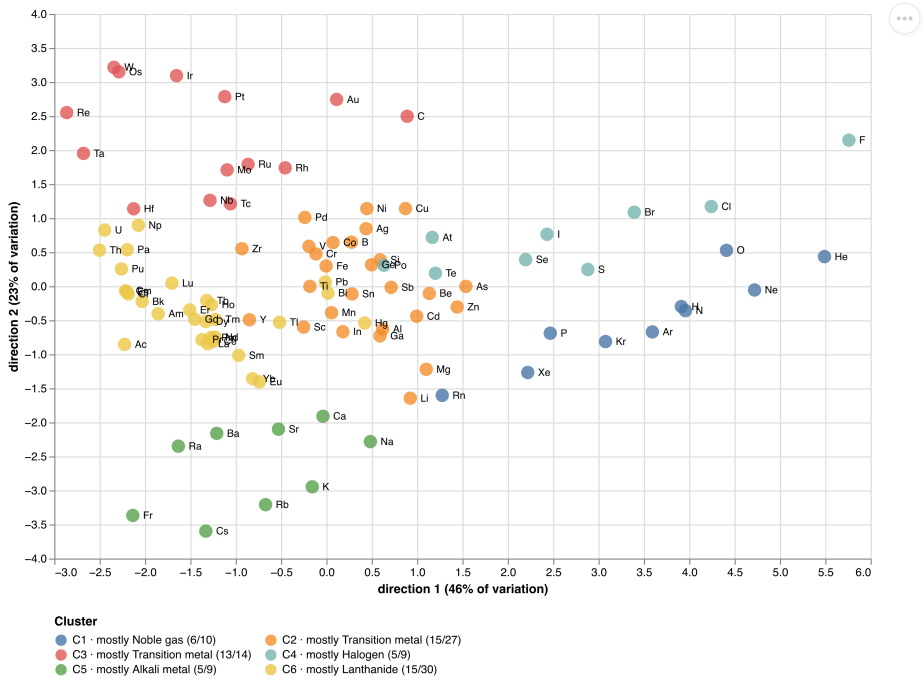
pairs	n pairs	observed same-cluster share	null mean	null 99.9th pct	p-value (one-sided)
all touching pairs	154	0.7338	0.2046	0.2987	0.0002
same group (up–down)	67	0.7164	0.2044	0.3731	0.0002
same period (left–right)	87	0.7471	0.2047	0.3333	0.0002



p = 1/5001 ≈ 0.0002 is the smallest value 5000 shuffles can give.

The clusters in the PCA map

This links Q1 and Q2. Each dot is an element, placed by the first two PCA directions from Q1. The clusters form separate areas of the map. **Scroll to zoom, drag to pan.**



Go deeper · Does the result depend on how we filled the blanks?

variant	n elements	ARI vs main clustering	neighbours in same cluster	chance level	p-value
fill blanks with: KNN, 3 neighbours	99	0.9748	0.7468	NaN	NaN
fill blanks with: KNN, 10 neighbours	99	1.0000	0.7338	NaN	NaN
fill blanks with: column median	99	0.9407	0.7468	NaN	NaN
only fully measured elements (no filled values)	55	0.2924	0.6711	0.1933	0.0002
20 other random seeds (mean)	99	0.9225	NaN	NaN	NaN

Other ways to fill the blanks give almost the same clusters (ARI near 1). The set of fully measured elements has no noble gases, no group-2 metals and almost no lanthanides, so its clusters are different (low ARI). But neighbours still share a cluster far more often than by chance. So the link with the table does not come from the filled-in values.

Answer to Q2

Yes. Elements that are close in the periodic table usually end up in the same k-means cluster, although k-means never saw the table.

- **73%** of neighbour pairs share a cluster. Chance gives **20%** ($p \approx 0.0002$).
- This is true **down a group** (72% vs 20%) and **along a period** (75% vs 20%).
- The clusters follow groups and blocks (ARI 0.42 with the group, 0.39 with the family) more than periods (ARI 0.25). Properties change fast along a period (metal → non-metal), but slowly down a group.
- The result stays the same when we fill the blanks in other ways, or use no filled values at all.

The 6 clusters are areas of the table

Cluster	Area of the table	Typical features
1	Noble gases + H, N, O, P	gases: highest ionization energy, lowest melting point and density
2	Light metals and semi-metals: Li, Be, Mg, Sc–Zn, Y, Zr, Pd, Ag, Cd, B, Al, Si, Ga, Ge, As, In, Sn, Sb	middle values
3	Middle of the heavy transition metals (Nb–Rh, Hf–Au) + C	highest melting point, boiling point and density
4	Halogens and chalcogens (F, Cl, Br, I, At, S, Se, Te, Po)	highest electron affinity
5	Groups 1–2 (Na–Fr, Ca–Ra)	lowest electronegativity and ionization energy, largest atoms
6	Lanthanides, actinides + Hg, Tl, Pb, Bi	heavy, large, low electronegativity

Where the features and the table disagree (each case makes sense chemically)

- **H** sits in group 1, but it is a gas. It joins the noble gases and N, O.
- **C** has the highest melting point of all elements. It joins the hard, high-melting metals.
- **Li, Be, Mg** are lighter and harder than the other group 1–2 metals. They join cluster 2.
- **Hg, Tl, Pb, Bi** are heavy and soft. They join the lanthanides and actinides.

Limit. With 6 clusters, k-means cannot rebuild 18 groups and 7 periods. It finds the big areas: the s-block edge, the d-block, the non-metals, the noble gases and the f-block.